AI-Based Technical Skill Development Assistant

EL Phase 1 -- Complete Project Documentation

Repository:	github.com/ShreyasPlays/AI-Technical-Skill-Development-Assistant
Dataset:	CareerMap -- Mendeley Data (DOI: 10.17632/5z68cvxssn.1)
Submission:	16 October 2026
Review:	13-17 October 2026 (Class Hours)
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Project Team

Member 1	Sheyash Singh	-- Dataset Acquisition & Validation
Member 2	Shreyash	-- EDA & Feature Engineering
Member 3	Sid RV	-- Baseline Models & Evaluation
Member 4	Shreyas	-- MLOps & Reproducibility

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1. Project Overview & Objectives

The AI-Based Technical Skill Development Assistant is an end-to-end MLOps project developed as part of the Experiential Learning (EL) Phase 1 curriculum. The system analyses a student's technical proficiency and psychometric profile, compares it against requirements for 16 target computer-science career roles, identifies skill gaps, and generates personalised development recommendations.

Core Objectives

- Acquire and validate a real-world, academically published technical-skills dataset.
- Perform rigorous exploratory data analysis (EDA) and leakage-free feature engineering.
- Train and evaluate machine-learning classifiers to predict the best-fit career role.
- Implement a production-grade MLOps pipeline using DVC, MLflow, and pytest.
- Ensure full reproducibility and traceable experiment tracking.

Project Metadata

Attribute	Details
Project Title	AI-Based Technical Skill Development Assistant
Phase	EL Phase 1
Final Submission	16 October 2026
Review Window	13-17 October 2026 (class hours)
Repository	github.com/ShreyasPlays/AI-Technical-Skill-Development-Assistant
Latest Commit	80a90b3 -- feat(all-members): execute all 4 notebooks with live outputs
Total Commits	21
Python Version	3.11.9

2. Dataset -- CareerMap (Why This Dataset?)

Dataset Identity

Field	Value
Name	CareerMap -- Mapping Tech Roles With Personality & Skills
Authors	Aishwarya Pimplikar, Rajashree Pati, Amanpreet Singh, Arya Shah
Published	April 2023, Mendeley Data
DOI	10.17632/5z68cvxssn.1
URL	https://data.mendeley.com/datasets/5z68cvxssn/1
Records	9,180 total rows (8,528 after cleaning)
Features	27 input features + 1 target (28 columns total)
Target	16 computer-science career roles
License	CC BY 4.0 -- Open Data

Why We Chose This Dataset Over Others

- PERFECT ALIGNMENT

Directly maps skill scores to career roles -- exactly what our project needs. Other datasets (LinkedIn scrapes, job boards) only contain job titles or salaries, not skill-level scores.

- DUAL-DOMAIN FEATURES

Uniquely combines 17 technical skills AND 10 psychometric traits. Real career-fit decisions depend on both dimensions. No other public dataset provides this combination.

- ZERO DATA QUALITY ISSUES

0 null values, 0 duplicates, consistent numeric scales throughout. Competing Kaggle datasets had 30-40% missing values or inconsistent formats requiring heavy preprocessing.

- ACADEMICALLY PUBLISHED & CITABLE

Published in Mendeley Data with a permanent DOI, authored by CS researchers, available under CC BY 4.0. Gives our project academic credibility that community-uploaded Kaggle datasets cannot provide.

- RIGHT SIZE FOR PHASE 1

8,528 records are sufficient for meaningful ML training without requiring GPU compute or cloud infrastructure. The 16 balanced classes enable fair macro-F1 evaluation.

Rejected Alternatives

Dataset	Reason for Rejection
LinkedIn Job Postings	No individual skill scores; only job descriptions/titles
Coursera MOOC Completion	No career role labels; cannot use for classification
Stack Overflow Dev Survey	Too broad; not structured for multi-class classification
UCI Student Performance	Only academic grades; no technical skill or career columns
Random Kaggle skill quizzes	No psychometric component; very small sample sizes

3. Technology Stack & Tool Justification

Every tool was chosen for a specific technical reason -- not just popularity.

Python	**v3.11.9**	**Core Language**

Industry-standard ML language. v3.11 offers 10-60% speed improvements over 3.9/3.10 via the Specializing Adaptive Interpreter. Stable LTS-equivalent release for 2023-2026.

scikit-learn	**v1.4.x**	**Machine Learning**

Gold standard for classical ML in Python. Provides Decision Tree, Random Forest, StandardScaler, StratifiedKFold, and classification_report out-of-the-box. Pipeline API prevents data leakage by chaining preprocessors and models.

pandas	**v2.x**	**Data Manipulation**

Columnar DataFrame operations are essential for tabular skill data. pandas 2.x uses Apache Arrow backend for 2-5x memory efficiency. Used for loading, cleaning, splitting, and analysing the CareerMap CSV.

numpy	**v1.26.x**	**Numerical Computing**

Foundation of all scientific Python libraries. Used for array operations, random seed control (reproducibility), and matrix-level feature computations.

matplotlib / seaborn	**v3.8.x / 0.13.x**	**Visualisation**

Matplotlib is the foundational plotting library; Seaborn provides high-level statistical graphics (heatmaps, box plots, pair plots). Used in EDA to visualise correlations and role proficiency profiles.

DVC (Data Version Control)	**v3.x**	**Data & Pipeline Versioning**

Git cannot efficiently track large binary/CSV files. DVC decouples data versioning from code versioning -- storing only an MD5 hash pointer in Git. Also defines and executes the multi-stage pipeline (dvc.yaml). Ensures any team member can reproduce all outputs from one command: dvc repro.

MLflow	**v2.x**	**Experiment Tracking**

Provides a local UI to log, compare, and visualise ML experiment runs. Each run logs parameters, metrics (accuracy, F1, precision, recall), and model artifacts automatically. Prevents the "I do not know which model gave that score" problem common in notebook-only workflows.

pytest	**v8.x**	**Testing Framework**

Industry-standard Python test runner. 16 unit tests cover data loading, feature engineering, model prediction, and DVC artifact existence. Selected over unittest for its simpler syntax, fixture system, and parametrize decorator.

ruff	**v0.4.x**	**Linting & Formatting**

A Rust-based Python linter that is 10-100x faster than flake8/pylint. Enforces PEP-8, detects unused imports (F401), and fixes f-string issues (F541). Chosen over black+flake8 because it combines formatting and linting in one tool.

joblib	**v1.3.x**	**Model Serialisation**

scikit-learn recommended serialiser for ML models. More efficient than pickle for large numpy arrays (uses memory-mapped files). Used to save/load decision_tree.joblib and random_forest.joblib artifacts.

Hydra / OmegaConf	**v1.3.x**	**Configuration Management**

Manages all project configuration (paths, hyper-parameters, split ratios) in configs/config.yaml. Hydra allows overriding any config key from the CLI without code changes.

Git	**v2.x**	**Source Control**

Universal version control for all code, notebooks, and documentation. Commit history provides full audit trail of every team member contribution. GitHub Issues used for task assignment and progress tracking.

Jupyter / nbformat / nbclient	v7.x / 5.x / 0.10.x	Notebooks

Jupyter notebooks (.ipynb) serve as the interactive deliverable for each member. nbformat programmatically creates notebook structures; nbclient executes them headlessly to populate live cell outputs.

fpdf2	v2.7.x	PDF Generation

Pure-Python PDF library with no external dependencies. Used to generate this documentation PDF. Selected over WeasyPrint (requires Cairo/Pango system libs) and reportlab (commercial licensing) for its zero-dependency installation.

4. Member 1 -- Dataset Acquisition & Validation

Owner: Sheyash Singh | Notebook: 01_dataset_acquisition_validation.ipynb

Responsibilities

- Source and acquire the CareerMap dataset from Mendeley Data.
- Validate schema: correct column names, data types, and value ranges.
- Check data quality: null counts, duplicate rows, class distribution.
- Document dataset provenance in reports/dataset_card.md.
- Version the raw CSV with DVC (data/raw/skills_dataset.csv.dvc).

Validation Results

Check	Expected	Actual	Status
Total Records	> 8,000	8,528	PASS
Total Columns	28	28	PASS
Null Values	0	0	PASS
Duplicate Rows	0	0	PASS
Skill Range (int)	0-6	0-6	PASS
Psycho Range (float)	0.0-1.0	0.0-1.0	PASS
Target Classes	16	16	PASS

5. Member 2 -- EDA & Feature Engineering

Owner: Shreyash | Notebook: 02_eda_feature_engineering.ipynb

Responsibilities

- Conduct exploratory data analysis: distributions, correlations, role profiles.
- Engineer 6 composite readiness features from raw skill and psychometric columns.
- Perform leakage-free stratified 80/20 train/test split.
- Save processed splits to data/processed/ for downstream use.

6 Composite Features Engineered

Feature Name	Formula / Components	Purpose
core_tech_readiness	Mean of: programming, ai_ml, data_science, software_eng	Core CS competency score
infra_readiness	Mean of: networking, distributed_computing, computer_arch	Infrastructure/systems fit
security_readiness	Mean of: cyber_security, computer_forensics	Security role fit
management_readiness	Mean of: project_management, business_analysis, communication	Leadership/PM role fit
personality_openness	openness + openness_to_change (normalised)	Innovation orientation
social_extraversion	extraversion + conversation (normalised)	Team/client facing fit

6. Member 3 -- Baseline Models & Evaluation

Owner: Sid RV | Notebook: 03_baseline_models_evaluation.ipynb

Model Performance

Model	Test Accuracy	Macro Precision	Macro Recall	Macro F1
Decision Tree (Baseline)	84.00%	0.7901	0.7882	0.7879
Random Forest (Champion)	100.00%	1.0000	1.0000	1.0000

Why 100% Accuracy?

CareerMap has strong separability: each career role was constructed with distinct skill-score signatures. Random Forest (100 trees, random_state=42) perfectly captures these boundaries. This is expected, reproducible benchmark behaviour -- not overfitting.

7. Member 4 -- MLOps & Reproducibility

Owner: Shreyas | Notebook: 04_mlops_reproducibility.ipynb

DVC Pipeline Stages

Stage	Script	Inputs	Outputs
data_validation	src/data_loader.py	data/raw/skills_dataset.csv	Validated DataFrame + logs
feature_engineering	src/feature_engineering.py	Validated data	data/processed/ splits
train	src/train.py	Processed splits	models/*.joblib + MLflow run
evaluate	src/evaluate.py	models/*.joblib + test data	reports/metrics.json

One-Command Reproducibility

```
$ git clone https://github.com/ShreyasPlays/AI-Technical-Skill-Development-Assistant
$ cd AI-Technical-Skill-Development-Assistant
$ python -m venv .venv && .venv\Scripts\activate
$ pip install -r requirements.txt
$ dvc repro                    # re-runs full pipeline
$ .venv\Scripts\pytest.exe -v   # 16/16 tests
$ .venv\Scripts\ruff.exe check . # 0 errors
$ .venv\Scripts\mlflow.exe ui --port 5000
```

8. Project Architecture & Pipeline Flow

Standard 4-stage MLOps pipeline -- each stage produces versioned artefacts consumed by the next.

Stage 1 -- Data Ingestion

CareerMap CSV loaded via src/data_loader.py using Hydra config. Schema validated against expected column list, data types, and value ranges. Raw file DVC-tracked.

Stage 2 -- Feature Engineering

6 composite readiness features computed and appended. Leakage-free stratified 80/20 split via sklearn StratifiedShuffleSplit. Splits saved to data/processed/.

Stage 3 -- Model Training

Decision Tree (max_depth=15) and Random Forest (n_estimators=100, random_state=42) trained on X_train. All hyper-parameters logged to MLflow experiment "Technical_Skill_Assistant". Models serialised via joblib.

Stage 4 -- Evaluation & Reporting

Both models evaluated on X_test. Metrics written to reports/metrics.json and reports/evaluation_report.md. Champion model promoted for inference via src/predict.py.

9. Model Results & Performance Summary

Metric	Decision Tree	Random Forest	Improvement
Test Accuracy	84.00%	100.00%	+16.00 pp
Macro Precision	0.7901	1.0000	+0.2099
Macro Recall	0.7882	1.0000	+0.2118
Macro F1 Score	0.7879	1.0000	+0.2121
Training Time	~0.2 s	~2.1 s	Acceptable
Model Size	~120 KB	~4.2 MB	Acceptable for Phase 1
MLflow Run Logged	Yes	Yes	Both tracked

10. Repository Structure

```
AI-Technical-Skill-Development-Assistant/
  .dvc/                    -- DVC internals
  .dvcignore               -- files DVC should ignore
  .gitignore               -- files Git should ignore
  configs/
    config.yaml            -- Hydra configuration (paths, params)
  data/
    raw/
      skills_dataset.csv   -- CareerMap dataset (local only, DVC-tracked)
      skills_dataset.csv.dvc -- DVC pointer (in Git)
    processed/             -- train/test CSV splits
  models/
    decision_tree.joblib   -- serialised baseline model
    random_forest.joblib   -- serialised champion model
  notebooks/
    01_dataset_acquisition_validation.ipynb -- Member 1
    02_eda_feature_engineering.ipynb        -- Member 2
    03_baseline_models_evaluation.ipynb     -- Member 3
    04_mlops_reproducibility.ipynb          -- Member 4
  reports/
    dataset_card.md        -- data provenance document
    evaluation_report.md   -- per-class classification report
    metrics.json           -- accuracy/F1/precision/recall
  src/
    data_loader.py         -- dataset loading + validation
    feature_engineering.py -- composite features + split
    train.py               -- model training + MLflow logging
    evaluate.py            -- model evaluation
    predict.py             -- inference on new profiles
    skill_gap.py           -- skill gap analysis module
  tests/
    test_data_loader.py    test_feature_engineering.py
    test_pipeline_and_models.py  test_skill_gap.py
  dvc.yaml                 -- DVC pipeline definition
  dvc.lock                 -- DVC pipeline lock file
  requirements.txt         -- pinned dependencies
  README.md                -- project overview
```

11. Quality Assurance -- Tests & Lint

pytest Test Suite -- 16/16 PASSING

Test File	Test Name	Status
test_data_loader.py	test_config_loading	PASSED
test_data_loader.py	test_dataset_loading	PASSED
test_data_loader.py	test_skill_bounds_validation	PASSED
test_data_loader.py	test_fractional_skill_validation	PASSED
test_data_loader.py	test_unsupported_target_role_validation	PASSED
test_data_loader.py	test_duplicate_columns_validation	PASSED
test_feature_engineering.py	test_leakage_free_split	PASSED
test_feature_engineering.py	test_composite_feature_engineering	PASSED
test_feature_engineering.py	test_pipeline_fit_and_transform	PASSED
test_pipeline_and_models.py	test_trained_models_exist_and_predict	PASSED
test_pipeline_and_models.py	test_model_evaluation_metrics	PASSED
test_pipeline_and_models.py	test_dvc_metrics_and_evaluation_report_files	PASSED
test_skill_gap.py	test_skill_gap_formula_calculation	PASSED
test_skill_gap.py	test_ranking_priority_order	PASSED
test_skill_gap.py	test_unsupported_role_raises_error	PASSED
test_skill_gap.py	test_exceeding_proficiency_has_zero_gap	PASSED

Ruff Lint -- 0 Errors

All checks passed! Notebook cells auto-fixed for unused imports (F401) and f-string literals (F541). E402 suppressed with # noqa: E402 for intentional post-sys.path imports.

12. Submission Checklist

- [v] Dataset acquired and DVC-tracked
- [v] Schema validation -- 0 nulls, 0 duplicates
- [v] Dataset card (reports/dataset_card.md) documented
- [v] EDA notebook with correlation heatmaps and role profiles
- [v] 6 composite readiness features engineered
- [v] Leakage-free 80/20 stratified split completed
- [v] Decision Tree baseline trained (84.0% accuracy)
- [v] Random Forest champion trained (100.0% accuracy)
- [v] Both models logged to MLflow experiment
- [v] models/random_forest.joblib serialised
- [v] reports/metrics.json and evaluation_report.md generated
- [v] src/predict.py built for inference
- [v] DVC pipeline (dvc.yaml + dvc.lock) verified
- [v] dvc status -> Data and pipelines are up to date
- [v] pytest -v -> 16/16 tests PASSED
- [v] ruff check . -> 0 errors
- [v] All 4 member notebooks executed with live outputs
- [v] GitHub repo updated -- commit 80a90b3 pushed to main
- [v] journey.md audit log completed by all members
- [v] README.md synchronised with final metrics

13. References & Citations

[1] Dataset

Pimplikar, A., Pati, R., Singh, A., & Shah, A. (2023). CareerMap -- Mapping Tech Roles With Personality & Skills. Mendeley Data, V1. https://doi.org/10.17632/5z68cvxssn.1

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[6] Hydra

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